Chapter 8. Conditional Independence and Gaussian Graphical Models

Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighborhood selection)

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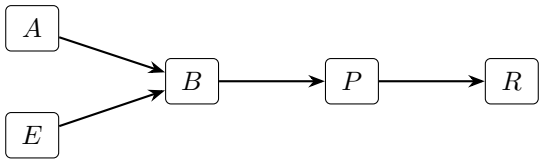
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Roadmap of the causal-discovery part

Lecture	Condition on the data	Relation to be discovered
1 (Ch. 8)	Gaussian observational data	undirected graph of conditional relations
2 (Ch. 9)	linear SEM with given directions	conditional independences a graph implies; limits of direction
3 (Ch. 10)	all main variables observed	the equivalence class of DAGs that data can distinguish
4 (Ch. 11)	hidden common causes allowed	latent confounding, relations and directions compatible with it
5 (Ch. 12)	extra assumption on the error distribution	more specific directions inside the equivalence class
6 (Ch. 13)	further identifiable models	directions for which causal discovery is possible

A running toy health example

The same five-variable health data are reused through all six lectures. The data-generating structure is



A	age	E	amount of exercise
B	body-mass index (BMI)	P	blood pressure
R	disease risk		

i.e. $E \to B \to P \to R$ and $A \to B$. Lecture 1 does not use directions and looks for conditional relations given the remaining variables; Lectures 2–3 study how directions are represented and how far they can be discovered; Lecture 4 adds a hidden lifestyle variable U, and Lecture 6 adds measurement error.

1 Correlation Mixes Direct and Indirect Relations

Let three standardized Gaussian variables be generated by

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ mutually independent, so all three variances equal $0.64 + 0.36 = 1$.

Pair	Correlation	Why the relation arises
X_1, X_2	0.8	X_1 and X_2 are directly connected
X_2, X_3	0.8	X_2 and X_3 are directly connected
X_1, X_3	0.64	the relation is transmitted through X_2

Indeed $\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The right question is not "is there a correlation?" but "**does a relation remain after accounting for the other variables?**"

2 Conditional Independence

Definition: conditional independence

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all z, equivalently $p(x \mid y, z) = p(x \mid z)$: among cases with the same Z, knowing Y gives no further information about X.

Marginal relation: $X_1 \not\perp\!\!\!\perp X_3$ — knowing X_1 changes what we know about X_3 (here Corr = 0.64). Conditional relation: $X_1 \perp\!\!\!\perp X_3 \mid X_2$ — comparing only cases with the same X_2, X_1 gives no additional information about X_3:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix causal direction, but it **narrows the candidate edges** of conditional dependence among variables.

3 Partial Correlation

Motivation. Two variables can be correlated only because both depend on a third. Partial correlation removes the linear effect of the other variables and asks whether a direct association remains; zero partial correlations define the edges missing from a Gaussian graphical model.

3.1 Definition by residuals

Adjusting X_1 and X_3 for X_2:

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

Quantity	Meaning
r_1	the part of X_1 not explained by X_2
r_3	the part of X_3 not explained by X_2
$\mathrm{Corr}(r_1, r_3)$	the partial correlation of X_1, X_3 given X_2

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals. For a set S, $\rho_{jk\cdot S}$ is defined in the same way after regressing both variables on X_S.

Theorem: first-order partial correlation

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Proof [Supplement]. Standardize all variables. The projection coefficients are $\beta_1 = \rho_{12}$, $\beta_3 = \rho_{23}$, so $r_1 = X_1 - \rho_{12}X_2$ and $r_3 = X_3 - \rho_{23}X_2$. Then $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$ and $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

Example: a correlation of 0.64 can disappear

With $\rho_{12} = 0.8$, $\rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0.$$

Under joint Gaussianity we therefore conclude $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

4 The Precision Matrix

4.1 Zeros of the precision matrix are conditional independences

Let $X = (X_1, \ldots, X_p)^\top \sim N_p(\mu, \Sigma)$ and call $\Omega = \Sigma^{-1}$ the **precision matrix**.

Precision matrix	Interpretation
$\Omega_{jk} = 0$	$X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$
$\Omega_{jk} \neq 0$	a relation between X_j, X_k remains given all the other variables

Theorem: precision matrix and full-order partial correlation

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

Proof [Supplement]. Let $A = \{j, k\}$ and B the rest. For a Gaussian vector, $X_A \mid X_B$ is Gaussian with covariance equal to the Schur complement $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$. By the block-inversion formula this Schur complement equals $(\Omega_{AA})^{-1}$, where Ω_{AA} is the 2×2 block of Ω. Hence

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

whose correlation is $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$. The conditional covariance does not depend on x_B, so this is the partial correlation. For a bivariate Gaussian, zero correlation is equivalent to independence. ■

One matrix inversion gives, for every pair simultaneously, the relation after accounting for all other variables.

4.2 Turning the precision matrix into a graph: the GGM

Definition: Gaussian graphical model

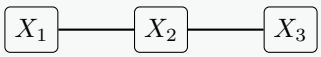
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \ldots, p\}$ with

$$(j, k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

Example: the three-variable chain

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$ (check $R\Omega = I$). So the GGM is



The conditional relations $X_1 - X_2$ and $X_2 - X_3$ remain; $\Omega_{13} = 0$ removes the edge $X_1 - X_3$. The partial correlation of X_1, X_2 given X_3 is $0.8/\sqrt{1.64} \approx 0.62$, smaller than the marginal 0.8 (the same value follows from the first-order formula).

Remark: from pairwise to global statements

The definition uses the *pairwise* Markov property. For a positive density (e.g. a nondegenerate Gaussian) it is equivalent to the *global* Markov property: if a set S separates node sets A and B in the graph (every path from A to B passes through S), then $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford). In the chain, $\{X_2\}$ separates X_1 from X_3.

5 Estimating the Graph

5.1 Inverting the sample covariance produces spurious edges

With the sample covariance S, the simplest estimate is $\widehat{\Omega} = S^{-1}$.

Data condition	Problem
finite sample	because of sampling error, entries that are truly 0 are estimated as small nonzero values
many variables	S^{-1} is unstable and most entries are nonzero
$p \geq n$	S is singular and the inverse may not exist

To discover relations we need a **sparse** precision-matrix estimate.

Method	Core idea	Suitable when
partial-correlation tests	compute partial correlations from S^{-1} and test $H_0 : \rho_{jk\cdot -\{j,k\}} = 0$	$p < n$ and sample large enough
graphical lasso	ℓ_1 penalty on the whole precision matrix, some entries estimated as 0	a sparse positive-definite precision matrix is needed
neighborhood selection	Lasso regression of each variable on the rest; choose neighbors variable by variable	many variables, node-wise relations
CLIME	directly estimate a sparse Ω satisfying $S\Omega \approx I$	estimating the precision matrix in high dimension

The edge definition of the GGM is the same; the estimation method depends on the data conditions.

5.2 Testing a zero partial correlation

For Gaussian data, $H_0 : X_j \perp\!\!\!\perp X_k \mid X_S \iff H_0 : \rho_{jk\cdot S} = 0$. Let $\hat{\rho} = \hat{\rho}_{jk\cdot S}$ be the sample partial correlation and $q = |S|$.

Test	Statistic	Reference distribution
partial-correlation t-test	$T = \hat{\rho}\sqrt{\dfrac{n-q-2}{1-\hat{\rho}^2}}$	t_{n-q-2}
Fisher-z test	$Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat{\rho})$	approximately $N(0,1)$
conditional distance correlation	$\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$	p-value by resampling

Derivation: where the degrees of freedom come from

Regressing on q variables plus an intercept uses $q+1$ degrees of freedom, so the residual correlation behaves like an ordinary sample correlation from $n-q$ observations. For an ordinary correlation from m pairs, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$ under $\rho = 0$, and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method with variance stabilization ($\mathrm{Var}(r) \approx (1-\rho^2)^2/m$ and $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$). Substituting $m = n-q$ gives the table.

Failing to reject H_0 does *not* prove conditional independence.

5.3 Graphical lasso

Derivation: Gaussian log-likelihood of Ω

For centered data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top\Omega x_i + c$, and $\sum_i x_i^\top\Omega x_i = \operatorname{tr}(\Omega\sum_i x_i x_i^\top) = n\operatorname{tr}(S\Omega)$. So the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \operatorname{tr}(S\Omega)\} + c$, maximized at $\Omega = S^{-1}$ when $S \succ 0$.

The graphical lasso combines Gaussian fit with sparsity:

$$\widehat{\Omega}_\lambda = \arg\min_{\Omega\succ 0}\Big[\operatorname{tr}(S\Omega) - \log\det(\Omega) + \lambda\sum_{j\neq k}|\Omega_{jk}|\Big].$$

Part of the objective	Role
$\operatorname{tr}(S\Omega) - \log\det(\Omega)$	matches the covariance structure of the observed data
$\lambda\sum_{j\neq k}\lvert\Omega_{jk}\rvert$	sets small off-diagonal entries exactly to 0

The objective is convex in Ω; its optimality condition is $\Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jk} \in \partial|\Omega_{jk}|$ (and $\Gamma_{jj} = 0$). Consequently $|(\widehat{\Omega}^{-1})_{jk} - S_{jk}| \le \lambda$ for all $j \neq k$, with equality on the estimated edges. The solution is positive definite even when $p \ge n$. λ is chosen by CV log-likelihood, EBIC or stability selection.

5.4 Neighborhood selection

Theorem: conditional distribution of one node

For centered $X \sim N_p(0,\Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}}X_{-j},\ \frac{1}{\Omega_{jj}}\Big),$$

so the population coefficients of the regression of X_j on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

Proof [Supplement]. Apply the Schur-complement argument with $A = \{j\}$: $\mathrm{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and the conditional mean $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}X_{-j}$ equals $-\Omega_{jj}^{-1}\Omega_{j,-j}X_{-j}$ by the block-inverse identity $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

In practice the population coefficients are estimated sparsely by the Lasso:

$$\hat{\beta}^{(j)} = \arg\min_{\beta} \Big\{ \frac{1}{2n}\|X_j - X_{-j}\beta\|_2^2 + \lambda\|\beta\|_1 \Big\}.$$

If $\hat{\beta}_k^{(j)} \neq 0$, $j - k$ is an edge candidate. In samples the two regressions (j on k and k on j) may disagree, so they are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one nonzero). Each regression costs one Lasso, which parallelises over nodes and scales to large p; the price is that the result is a graph, not a positive-definite $\widehat{\Omega}$.

5.5 CLIME

$$\widehat{\Omega} = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \leq \lambda,$$

a linear program that splits into one column at a time; the estimate is symmetrized afterwards. It estimates the sparsest matrix that is approximately an inverse of S.

6 GGM Edges Are Conditional Associations, Not Causal Directions

Interpretable from a GGM	Not interpretable from a GGM alone
Does a relation remain after accounting for the other observed variables?	Which variable is the cause and which the effect?
What are the sign and size of the partial correlation?	Is an edge a direct causal effect?
Which variables are conditionally connected to many others?	Is there a hidden common cause?

For example, $X_j - X_k$ means that the two variables remain related after accounting for the remaining observed variables. It means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$. Worse, conditioning on a common *effect* can create an edge between two independent causes (Ch. 9). To infer directions we need DAGs and additional structure-learning assumptions.

Summary

- Correlation mixes direct and indirect relations; conditional independence asks what remains given the other variables.
- Partial correlations are correlations of residuals; for Gaussians, $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ and $\rho_{jk\cdot\mathrm{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$.
- The GGM puts an edge where $\Omega_{jk} \neq 0$; it is estimated by partial-correlation tests, graphical lasso, neighborhood selection or CLIME.
- GGM edges are undirected conditional associations; directions require the DAG machinery of the next chapters.